Statement of Purpose

elhanasif8@gmail.com

University of California, Irvine | PhD in Software Engineering Md. Asif Haider | asifhaider.github.io

Software systems embody the technical solutions to almost every practical challenge in today's world. The overarching research problem that intrigues me the most is how to improve the quality of these software systems: here quality software refers to more reliable, secure, and efficient systems. These systems are, after all, built by humans, for the use of humans- leaving significant human factors and developer perspectives in play as well. As such, my academic and research background has led me to identify two important dimensions of my research interest: 1) automated testing, fault localization, and repair of software systems to increase reliability, and 2) development and evaluation of software tools/techniques ensuring efficiency and developer productivity. Recent advances in Large Language Models (LLM) have opened up remarkable opportunities to address such problems, also introducing newer challenges. My primary motivation for pursuing a PhD is to get training on how to design and perform scientific experiments to explore such research ideas independently on my own, sharpening my knowledge of software engineering, machine learning, and human factors in computing. I believe that four years of my undergrad experience in research publications, and international collaborations, coupled with my solid academic background and diverse skillset, make me a fit for the Software Engineering PhD program at the esteemed University of California (UC) Irvine.

I first came to know about software automation while taking the Software Engineering course in my junior year. I took part in (and won) a couple of DevOps hackathons that fueled my interest in automating software pipelines. GitHub Copliot also came out around that time, underlining the potential of LLMs in advancing code-related tasks (LLM4SE), a direction I was looking forward to exploring. In early 2023, I had one such opportunity to work under Prof. Joanna Santos (University of Notre Dame) as a visiting student researcher. Our study investigates the effectiveness of Codex in zero-shot code complexity prediction task, where it was found to outperform other popular language models based on BERT, T5, and BART for linear complexity class problems. Our findings were presented at the NLBSE Workshop @ ICSE 2023 conference. In addition to experimenting with language models, here I contributed to data collection, cleaning, and visualization. The success of this preliminary study inspired me to choose similar topics for my further research.

Automating code review activities poses a significant demand, as it takes a huge toll on the developer's valuable time and effort. How can we leverage LLMs to improve automation performance here? Driven by this question, I worked with Prof. Anindya Iqbal (BUET) and Dr. Toufique Ahmed (IBM Research) in my undergraduate thesis focusing on two code review tasks: review comment generation and code repair generation from given code patch. While large-scale pre-training is promising, it can hardly capture project-specific nuances while requiring substantial training data and computing resources- leaving a negative carbon footprint. To tackle this problem, first I adopted a parameter-efficient fine-tuning (PEFT) strategy QLoRA to fine-tune open-source Llama models, significantly reducing the memory requirement while maintaining an acceptable accuracy for our tasks. However, we can do better in the era of prompt engineering! Could semantic program metadata improve the prompt quality? Our novel contribution includes appending CodeT5-generated natural language summary and statically extracted function call graph into the LLM prompts. Our BM-25 powered, callgraph-augmented few-shot prompts with GPT achieved a 90% BLEU-4 improvement in generating review, while Gemini reached 97% of the baseline performance in generating repair. My first-authored manuscript from this project is currently under review at the reputed MSR 2025 conference. Leading the project, I also crafted a detailed research proposal which eventually won us a research grant worth 800+ USD. I consider this research work a very rewarding part of my journey, learning to tackle different challenges (maintaining large codebases, tracking progress via documentation, managing funds) while actively contributing to static analysis, LLM fine-tuning, and prompting.

While studying how to improve code repair performance, I was drawn to another related problem: code generation and debugging, specifically, finding out where LLM-generated code fails and why. Under the guidance of Prof. Eunus Ali (BUET) and Dr. Rizwan Parvez (QCRI @ HBKU), next, I focused on solving competitive algorithmic problems powered by automated, iterative debugging. Building upon a recent work, we adapt prompting open-source LLM-backed multi-agent retrieval, planning, and debugging. Our experimental

results show chain-of-thought few-shot prompting of DeepSeek Coder and Codestral surpasses the performance of ChatGPT, while Qwen-2.5 Coder outperforms even GPT-4. My current plan includes developing a detailed error taxonomy of LLM-generated codes and consequently researching an efficient way to repair them. A few months before my graduation, I also had the opportunity to intern at a reputed software security service provider company, OpenRefactory Bangladesh Limited, where I worked with their proprietary SAST tool, Intelligent Code Repair (iCR). My task was to triage iCR-generated fix hints to locate Python and Java security bugs in Open Source Software (OSS) projects, manually fix them, and communicate with OSS maintainers through detailed documentation and GitHub pull requests.

All these experiences ignited my interest in studying automated program analysis and repair, software quality assurance, and developer education and productivity- topics I would love to explore throughout my PhD. My current interests align most closely with Prof. Daye Nam. I had the privilege to meet with her to discuss a potential fit. Under her guidance, I would love to apply natural language processing and program analysis techniques to build more productive and usable tools for the developers. Specifically, given my prior experience in Code LLMs and software data mining, I hope to contribute to code understanding and contextual explanation generation ideas. I would also be interested in working with Prof. Iftekhar Ahmed and Prof. Mohammad Moshirpour, as their works on AI for code comment consistency, commit message generation, bug and mutation analysis tasks highly align with my prior works. Under any of their supervision, I would be happy to dive deep into similar directions. With an additional interest in software testing and accessibility research, I would also be happy to work with Prof. Sam Malek, focusing on android test automation, accessibility test report and test oracle generation- some ideas to start with. Given my proven research background with the ability to grasp diverse contents quickly, I hope to be a successful contributor to any of their groups.

Soon after my graduation, I joined Brac University as a Lecturer in Computer Science driven by my passion for teaching and mentoring. I envision myself as a highly motivated, self-driven individual building a successful career as a professor/scientist working in my own research lab, driven by the motivation to ensure social good through my work. I believe a PhD degree will be a stepping stone for my goal to be a leader in world-class research, upholding my little country on the global landscape. The Information and Computer Sciences department at UC Irvine stands out as an obvious choice for my PhD studies due to its excellent contribution to Software Engineering, Artificial Intelligence, and HCI research. Post-PhD, I would love to see myself contributing back to the community through teaching and impactful research, and I believe the unparalleled resources and networking opportunities at UC Irvine will help me turn my aspirations into reality.

[1200 Words]